Multiscale Attention Fusion Graph Network for Remote Sensing Building Change Detection

Yu Shangguan, Jinjiang Li, Zheng Chen, Lu Ren, and Zhen Hua

***Abstract*— With the development of imaging systems and satellite technology, higher quality high-resolution remote sensing (RS) images are being applied in building change detection (BCD) techniques. Methods based on convolutional neural network (CNN) have achieved excellent success in BCD techniques due to their excellent feature discrimination ability. However, CNN relies heavily on the geometry of prior conditions and is limited by the size of the convolution kernel, making it easy to ignore global information. This makes it difficult to capture the long-range dependence of different building targets and handle complex spatial relationships in high-resolution satellite RS images. Considering that graph convolutional neural networks (GCNs) have powerful internal relationship learning capabilities, we propose a multiscale attention fusion graph network (MAFGNet) in this article. MAFGNet uses a dual graph convolution module (DGM), which includes a spatial graph convolution network (SGCN) and a channel graph convolution network (CGCN), to effectively explore the long-range relationship between the detection target and the global at the spatial and channel levels. We also design a multiscale attention fusion encoder that includes channel attention fusion module (CAFM) and spatial attention fusion module (SAFM) to effectively combine valuable information from multiscale features. In addition, an atrous context self-attention pyramid (ACSP) is designed to combine multiscale context to enhance the feature representation of change information. We conducted qualitative and quantitative comparative experiments on different datasets to validate the effectiveness of our model. The experimental results show that our method performs better than advanced methods in terms of overall accuracy (OA) and visualization details. Our code is available at https://github.com/ShangGY805/MAFG.**

***Index Terms*— Atrous context self-attention pyramid (ACSP), attention fusion module (AFM), building change detection (BCD), dual graph-convolution module, high-resolution remote sensing (RS) images.**

I. Introduction

A BUILDING change detection (BCD) technique applied to remote sensing (RS) images is defined as a technology that identifies and extracts the change information of building entities between a pair of registered multitemporal images [1]. In recent years, RS datasets have become increasingly high-resolution, with very high-resolution (VRH) RS images containing more detailed and richer information. Advanced CD technology has outstanding analytical and processing capabilities for change information in multitemporal remote sensing images, effectively reducing a large amount of manual labor and time costs. Therefore, it plays an increasingly important role and receives widespread attention in the fields of regional disaster assessment, urban planning, environmental detection, and map information updating [2], [3], [4], [5], [6], [7]. Fig. 1 shows the RS images of the same area at different times and the result of BCD. The image uses binary labels to distinguish the change type of each pixel [7], where positive labels indicate that the pixel has changed, and negative labels indicate that there is no change.

In the past period of time, BCD technology has been continuously improving to meet the demand for higher quality RS image tasks. Currently, convolutional neural networks (CNNs) are widely used in various computer vision tasks and have achieved great success due to their ability to capture rich feature information and better representation for complex visual tasks. CNN-based BCD methods have also shown good performance [8], [9], [10], [11], [12], [13]. As ResNet [14] can effectively increase network depth and learn richer deep semantic information, it is often used as the backbone network of advanced BCD networks to extract multilevel features of dual-time RS images [15]. The UNet structure proposed by Ronneberger et al. [16] is also widely used in BCD tasks [17], [18], [19], [20]. It adopts an encoder–decoder structure and uses cross-layer connections to transfer low-level information to high-level ones to enhance the model's contextual modeling ability. Many works have used dilated convolution [21] and attention mechanisms [13], [22], [23], [24], [25], [26], [27] to increase the model's receptive field and obtain rich global information, focusing on the interested change features. Although these methods achieve good performance, there are still some problems.

1) Many existing methods are accustomed to using stacked convolution and transformer-style operations to extract features [23], [28], [29], [30], [31]. This is not only inefficient but also does not fully utilize the correlation of multiscale features, thereby limiting the model's effective extraction of features.
2) Many existing works use simple fusion operations as the final step to obtain change results without more further processing [8], [23], [28], [31], [32],

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Yu Shangguan, Jinjiang Li, Zheng Chen, and Lu Ren are with the School of Computer Science and Technology, Institute of Network Technology, Shandong Technology and Business University, Yantai 264005, China (e-mail: chenzheng@sdtbu.edu.cn).

Zhen Hua is with the School of Computer Science and Technology, Shandong Technology and Business University, Yantai 264005, China.

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Fig. 1. Prechange image (left), postchange image (center), and ground truth (right) on LEVIR-CD [8], where the yellow box marks the location of the change target.

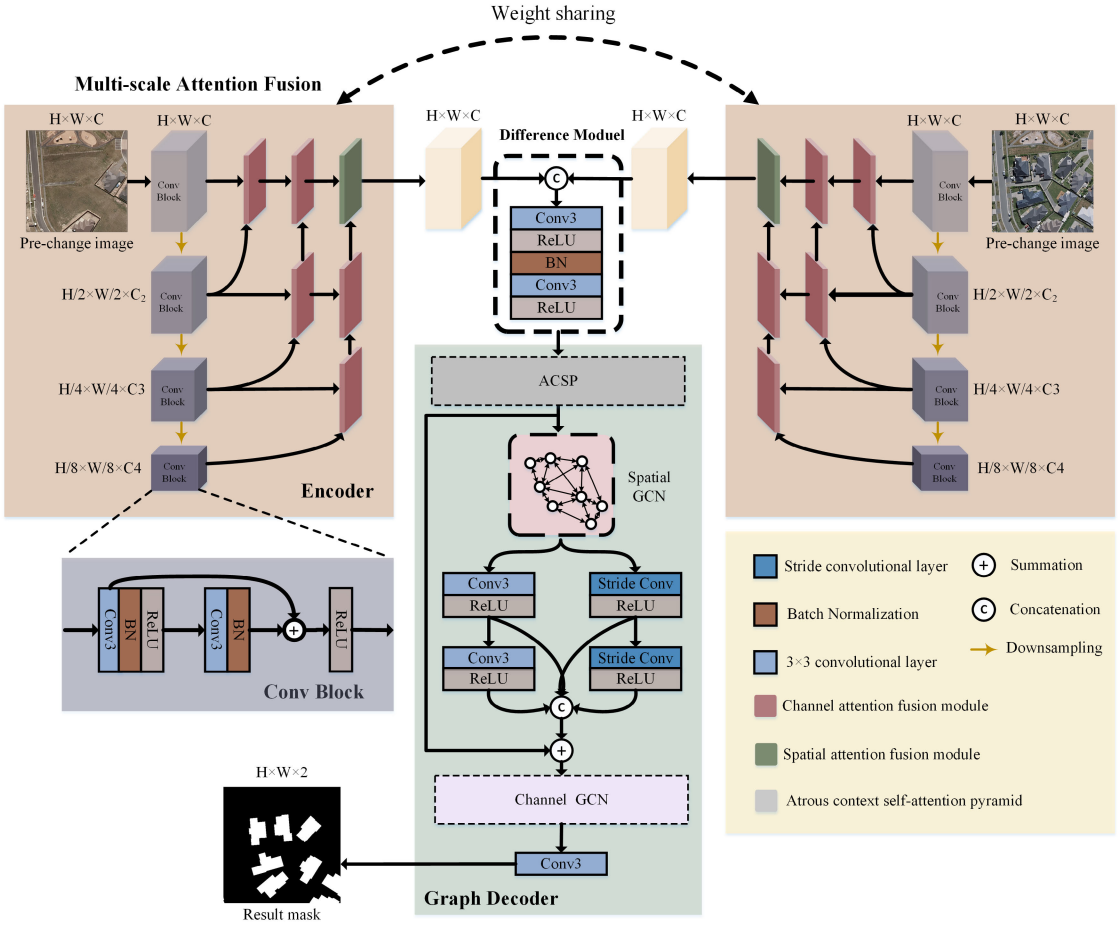


Fig. 2. Overall architecture of MAFGNet. The encoder part contains CAFM and SAFM to combine multiscale features. The decoder part obtains the change discrimination feature through the difference module and then uses two different graph convolution modules and the ACSP module to strengthen the spatial context modeling ability of the model.

which ignores further exploration of effective change information.

3) High-resolution RS images reflect more complex spatial and spectral characteristics, especially the relationship between the changing target of interest and other targets, which makes the long-range dependence between different regions critical.

In order to effectively solve the above problems, we propose a multiscale attention fusion graph network (MAFGNet), as shown in Fig. 2, to achieve effective BCD of RS images. MAFGNet uses attention fusion to combine multiscale global and local feature information and then enriches the change feature representation through the proposed ACSP and two graph convolutions under different mapping spaces to obtain an effective CD result.

The contributions of our proposed MAFGNet are summarized as follows.

1) An attention-based multiscale feature fusion structure is designed to progressively fuse multiscale features, which is more efficient than ordinary element-level fusion

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methods. Spatial and channel attention plug-ins are used to retain valuable spectral information in high-level features and spatial detail information in low-level features to enhance the recognition of building objects and improve their edge information.

2) In order to effectively model the long-range interdependence between different building objects in the change feature, a dual graph convolution module (DGM) is introduced. DGM combines a spatial graph convolution network (SGCN) and a channel graph convolution network (CGCN) to capture long-range contextual information in different mapping spaces, which fully utilizes the advantages of graph convolutional neural network (GCN) in integrating global information to significantly enhance the feature representation of change information.
3) We propose an ACSP to explore multiscale contextual information of the coarse change feature. ACSP utilizes context self-attention (CSA) branches with different dilated rates to learn multiscale semantics of the change feature, thus achieving adaptive perception and capturing of building targets of various sizes.

The remainder of this article is organized as follows. The related work in Section II presents some deep-learning CD method and graph convolutional network models. The overall network structure and implementation principles of different modules of MAFGNet are described in Section III. Section IV describes the experimental part, including the datasets, evaluation metrics, and experimental results. The conclusion is given in Section V.

II. Related Work

A. Deep Learning-Based CD Techniques

The method based on deep learning, which combines multiple processing layers to automatically acquire abstract feature representations at different levels, has been widely applied in the field of RS image analysis [7]. In the application of CD in dual-temporal RS images, many advanced deep learning methods have demonstrated excellent performance. These methods can be categorized as supervised and unsupervised methods depending on the availability of labeled training data for the task.

In the past, there were often not enough labeled training data and ground-truth information that can indicate the changes between two different time points of RS images, which supported the development of unsupervised methods for learning change features of dual-temporal images with broader applicability [33]. To achieve the distinction between change regions and irrelevant regions, an unsupervised method proposed by Su et al. [34] first used a deep belief network (DBN) to transform the spectral channels in the input image into abstract feature space. Then, the feature change analysis (FCA) was applied to discriminate the change information. Su et al. [34] proposed a new unsupervised deep learning framework to tackle the information-imbalanced ternary CD task. This framework first processed the input images through stacked denoising autoencoders and used the mapping functions generated by stacked mapping networks to reflect the relationships between different types of features. After comparing these features, clustering is performed on the comparison results to generate the final triplet map. Li et al. [35] trained their deep network using the CD maps from the existing unsupervised CD methods to eliminate noise. This network is an end-to-end, noise-modeling-based unsupervised fully convolutional network (FCN) framework mainly applied to CD in hyperspectral images. Huang et al. [36] proposed a new unsupervised algorithm to detect building changes in high-resolution multitemporal images. This algorithm first used convolutional operations to extract low-level features, such as texture and spectral features, and generated a combined low-level feature vector for each pixel. DBN was used in this algorithm for pretraining unlabeled samples. Afterward, the parameters were optimized through the extreme learning machine (ELM) classifier [37]. Considering the challenges brought by seasonal changes and style differences of multitemporal images to CD, Xu et al. [38] studied a CD with a domain shift setting and, based on this, designed a new unsupervised CD method using a lightweight transformer. This method uses a transformer-driven image translation to produce visually plausible stylization images, which can mitigate domain shifts between two images and produce an efficient difference maps with real-time efficiency. In addition, Wu et al. [39] also proposed an end-to-end fully convolutional framework with a generative adversarial network to achieve the induction of unsupervised, weakly supervised, regionally supervised, and fully supervised CD tasks.

Compared with unsupervised methods, supervised deep learning methods often demonstrate superior performance with the support of a large amount of applicable training samples and ground truth and have become a hot research topic. The purely supervised CNN model maps the input image into feature maps of different levels, which contain distinguishable information for bitemporal images. The UNet network proposed by Ronneberger et al. [16] has achieved excellent performance in the task of medical image segmentation. The network uses a symmetrical encoder and decoder to further transmit multilevel feature information and is used in CD tasks as a classic structure and has been continuously improved and expanded. Jaturapitpornchai et al. [40] proposed a UNet-based network for BCD tasks using synthetic aperture radar (SAR) images from two different times. This network uses an encoder to extract features and a decoder to segment the change result image. Based on the UNet structure, Peng et al. [41] proposed an improved UNet++ method that uses dense connections between extracted features of different scales to achieve end-to-end CD detection for high-resolution RS images. Fang et al. [42] proposed a dense connection Siamese network to effectively reduce the loss of low-level feature information and deep localization information. Attention mechanisms are frequently used in supervised learning methods to obtain effective information from the extracted features and enhance the model's directional learning ability. Attention mechanisms include channel attention [13], [22], [23], [25], spatial attention [22], [23], and self-attention [11], [26], [27]. Chen and Shi [8] proposed a Siamese network based

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on the fully convolutional neural network, which combines spatial and channel attention to obtain more change feature information. Zheng et al. [43] proposed a high-frequency spatial attention network that focuses on changing buildings and enhances the model's perception of high-frequency information for change buildings, thereby further improving detection accuracy. The vision transformer (ViT) [44], which is improved from the transformer used for natural language processing (NLP), has also been shown to exhibit excellent performance in CD tasks [28], [45]. It divides the input image into multiple image patches and uses multihead self-attention to learn the relationship between each image patch and other image patches. Chen et al. [28] used the transformer to design a method for CD in RS images, abstracting the interested change areas as visual words, i.e., semantic tokens. These tokens are modeled by the transformer encoder to obtain rich contextual information, and then they are fed back to the pixel space through the transformer decoder to refine the original features. Bandara and Patel [45] combined multiscale transformer encoders and multilayer perceptron (MLP) decoders to meet the multiscale remote details required for high-precision CD tasks. In addition, in order to effectively extract local and global features, Zhang et at. [46] proposed an asymmetric cross-attention hierarchical network that combines CNN and transformer in a series–parallel manner, which enhances the interaction between CNN and transformer.

This article adopts a multiscale learning strategy in the encoder part to capture feature information of different resolutions to effectively distinguish different sizes of change targets. We also introduce a attention fusion module (AFM) to improve the efficiency of feature fusion and focus on useful feature content as much as possible.

B. Graph Convolutional Neural Networks

The majority of data structure relationships in reality are expressed as complex non-Euclidean space relationships, which make the translational invariance of Euclidean space no longer applicable and, thus, make it difficult for traditional CNN to model them. Bruna et al. [47] first proposed the GCN, which is based on graph theory and combines the convolution defined in spectral space to define graph convolution, which later developed into the spectral method of the GCN field. The GCN can effectively analyze and process complex graph structured data, and provide reasonable explanations for the relationships between nodes and edges in the graph structure. Compared with CNN, the GCN does not depend on prior geometric shapes but maps the research objects to nodes, and the correlations between them are regarded as edges. Through convolution operations, it integrates spatial data and extracts higher level features from the graph structure data. The GCN can also learn the implicit internal relationships between objects in the input data. GCN has recently been widely applied to computer vision tasks [48], [49], [50], [51], [52]. Li and Gupta [53] proposed a method that converts 2-D images into graph structures in the semantic segmentation task, where different pixel clusters are defined as nodes in the graph, and the similarity between these clusters in the feature space represents the edges of the graph. The method then remaps the learned graph representation back to the 2-D grid space to capture long-range dependencies between different regions for higher accuracy segmentation. Zhang et al. [54] proposed a dual GCN framework to model global context for semantic segmentation. One branch of this framework is used to model the mutual dependencies between pixels in the spatial space of the input image, while the other branch is used to model the mutual relationships on the channel level. Qu et al. [55] first introduced GCN in the CD task of hyperspectral images to identify differences between multitemporal images. In their method, they proposed a novel dual-branch differential amplification GCN to extract different features in the image and amplify the difference features between them for effective CD. In addition, Xu et al. [56] used a contour-graph reasoning module for CD of RS images to capture the semantic relationship between the target area and the contour to improve the edge information.

III. Methodology

In this section, we first introduce the overall framework of our network. Then, we describe three main components in the model in detail: DGM, AFM, and ACSP. Finally, we illustrate the loss function and feature difference module and summarize the entire training process of the network in an algorithmic form.

A. Overall Framework of the Network

The model based on CNN has shown good performance in CD tasks, but it tends to focus on local feature data, which also limits the connection between regions in the detected image. In order to effectively explore the long-range dependencies between regions, a GCN focusing on the study of one-to-many data relationships is applied to our network structure. The overall structure of the network we designed is shown in Fig. 2. To better learn shallow information, such as edge and shape detection of target objects as well as deep semantic information from the input bitemporal images during network training, we have designed a Siamese multiscale feature attention fusion structure in the encoder section of the network. This process of feature refinement is weight-shared for bitemporal input images. The channel AFM (CAFM) and spatial AFM (SAFM) will be used to combine those intermediate features with different resolutions more efficiently, which will effectively explore multiscale feature information. After that, we combine the acquired intermediate refinement features through a simple difference module and obtain a change-discriminating feature. This change-distinguishing feature is still relatively rough. We then use ACSP to combine the multiscale contextual information of the change feature and use them to guide more effective self-attention learning. We further map the change feature to the graph domain in the decoder and learn the long-range correlation between different node features in the graph structure at the spatial and channel dimensions through SGCN and CGCN to enhance the detection performance of the model.

The procedure of our method can be illustrated by Algorithm 1.

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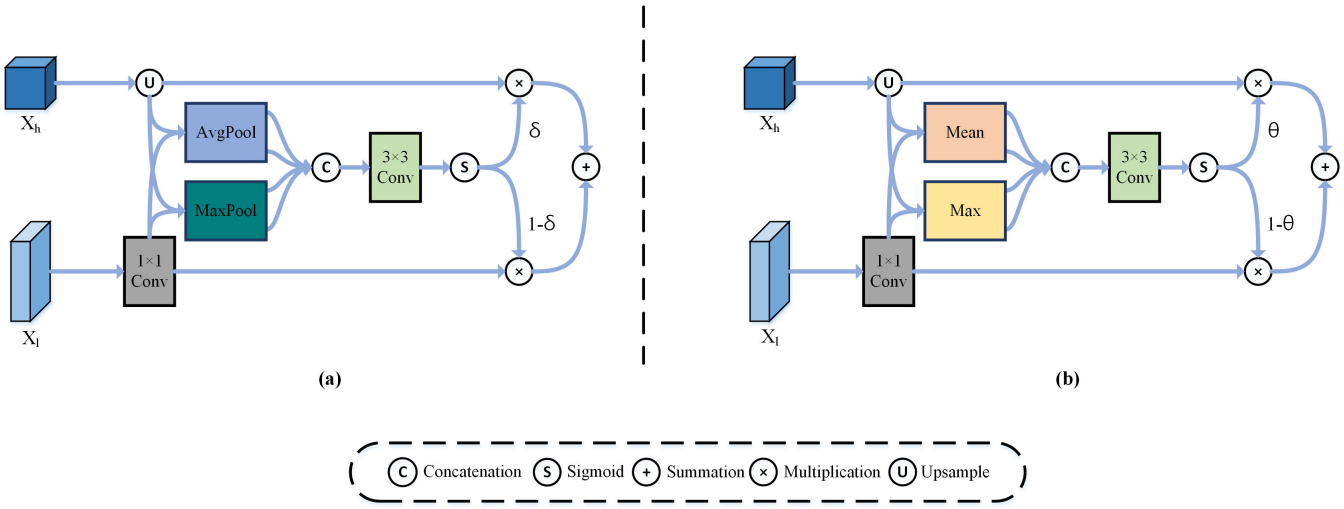


Fig. 3. Description of the AFM. (a) Channel attention module. (b) Spatial attention module.

Algorithm 1 Implementation Steps of MAFGNet Model

Input: $\mathbf{X_1}, \mathbf{X_2}$(*A pair of satellite remote sensing images at different times*)
Output: O
// *step*1 : *Use the attention fusion module to fuse multi − scale features in the encoder.*
for i *in* $\{1, 2\}$ **do**
  for j *in* $\{1, 2, 3, 4\}$ **do**
    $\mathbf{X_{i,j}} = ConvBlock(\mathbf{X_i})$
  end
  for k *in* $\{3, 2, 1\}$ **do**
    $\mathbf{X'_{i,k}} = CAFM(\mathbf{X_{i,k+1}}, \mathbf{X_{i,k}})$
  end
  for m *in* $\{2, 1\}$ **do**
    $\mathbf{X''_{i,m}} = CAFM(\mathbf{X'_{i,m+1}}, \mathbf{X'_{i,m}})$
  end
  $\mathbf{X''_i} = SAFM(\mathbf{X''_{i,1}}, \mathbf{X''_{i,2}})$
end
// *step*2 : *Calculate the feature difference*
$X_d = Difference_Module(X''_1, X''_2)$
//*step*3 : *Use ACSP to further enhance feature representation*
$\mathbf{X_a} = ACSP(\mathbf{X_g})$
// *step*4 : *Exploiting internal interrelationships through dual − graph convolutional*
$\mathbf{X_s} = SGCN(\mathbf{X_d})$
$\mathbf{X'_s} = Concat(Conv(\mathbf{X_s}), StridConv(\mathbf{X_s}), \mathbf{X_s}) + \mathbf{X_d})$
$\mathbf{X_g} = CGCN(\mathbf{X'_s})$
//*step*5 : *obtain the result of binary mask*
$\mathbf{O} = Conv(\mathbf{X_a})$

B. Attention Fusion Module

Effectively fusing features from different scales is the key to multiscale learning. Concatenating operations and element-level summation are conventional methods for fusing multiscale features. However, these methods lack more efficient means to explore valuable information of different scale features during the fusion process. To strengthen the effectiveness of our network in combining features of different resolutions, we introduce a unified AFM, as shown in Fig. 3, which provides different options of spatial and channel attention. The use of the attention mechanism makes the fusion process more efficient, and the obtained reweighted fused features are richer in information. Fig. 4 shows the designed multiscale attention fusion process based on AFM, and $L1$–$L4$ represent different fusion layer numbers.

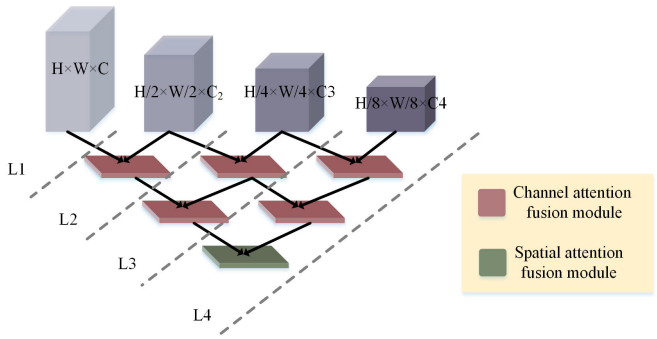


Fig. 4. Process of multiscale feature attention fusion. $L1$–$L4$ represent different levels of structure.

As shown in Fig. 3, when features of two different scales are input to AFM, they need to be preprocessed first. Specifically, given inputs $X_l \in \mathbb{R}^{C\times H\times W}$ and $X_h \in \mathbb{R}^{C'\times H/r\times W/r}$, where C represents the number of channels of low-level features, C' represents the number of channels of high-level features, H and W represent height and width, respectively, and r represents the multiple of upsampling. The high-level feature X_h with a smaller size is first upsampled to $X_{h'} \in \mathbb{R}^{C'\times H\times W}$ by bilinear interpolation operation, while the low-level feature X_l is changed to $X_{l'} \in \mathbb{R}^{C'\times H\times W}$ with the same channel number as the high-level feature through a 1×1 convolutional layer. This process can be expressed as follows:

$$\begin{aligned} X'_h &= \text{Upsample}(X_h) \\ X'_l &= \text{Conv1} \times 1(X_l). \end{aligned} \tag{1}$$

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As shown in Fig. 3(a), the CAFM uses a channel attention plugin to fuse the input features and generate a weight of δ to reweight them. This weight shows the respective importance of different channels in the input features. Specifically, we perform average-pooling and max-pooling operations on the two features X'_h and X'_l obtained by preprocessing to obtain four features with a dimension of $R^{C'\times 1\times 1}$. These four features are concatenated along the channel axis and rescaled to channel number C' using convolution operations. Then, we use the sigmoid function to generate a weight δ with a dimension of $R^{C'\times 1\times 1}$ and perform element-level multiplication with the two input features, respectively, and finally sum them to obtain the fusion feature. This process can be summarized in the formula

$$\begin{aligned} X_c &= \text{Concat}\big(\text{AvgPool}\big(X'_h\big), \text{MaxPool}\big(X'_h\big) \\ &\quad \text{AvgPool}\big(X'_l\big), \text{MaxPool}\big(X'_l\big)\big) \\ \delta &= \text{Sigmoid}(\text{Conv3}\times 3(X_c)) \\ X_{\text{out}} &= X'_h\cdot\delta + X'_l\cdot(1-\delta). \end{aligned} \tag{2}$$

As shown in Fig. 3(b), similarly, the SAFM uses a spatial attention plug-in to generate weight θ, which reveals the importance of different pixels in the input features. Specifically, we perform max and mean operations on the two preprocessed features of the input along the channel dimension to obtain four compressed features with a dimension of $R^{1\times H\times W}$. After these four features are concatenated and convoluted, sigmoid is used to generate weight θ with dimension $R^{1\times H\times W}$. Finally, we use to perform element-level multiplication with the two preprocessed features and then add them. This process can be described by the formula

$$\begin{aligned} X_c &= \text{Concat}\big(\text{Mean}\big(X'_h\big), \text{Max}\big(X'_h\big) \\ &\quad \text{Mean}\big(X'_l\big), \text{Max}\big(X'_l\big)\big) \\ \theta &= \text{Sigmoid}(\text{Conv3}\times 3(X_c)) \\ X_{\text{out}} &= X'_h\cdot\theta + X'_l\cdot(1-\theta). \end{aligned} \tag{3}$$

C. Dual-Graph Convolutional Network Module

To efficiently explore long-range spatial and channel relationships in features containing varying information, we design a dual graph convolutional network module. This module uses two different graph convolutional network modules, including the SGCN module and the CGCN module.

1) SGCN Module: For the task of BCD in bitemporal images for optical RS, it is crucial to be able to effectively distinguish between changing and nonchanging objects. Spatial graph convolution fully explores the long-range contextual information between different objects at the spatial level by aggregating global pixels and modeling the internal interrelationships between individual pixels and all pixels, which can enhance the sensitivity of the network to changing objects and, thus, improve the detection performance. Therefore, we use SGCN to capture local-to-global spatial information. Specifically, given a feature $X\in\mathbb{R}^{C\times H\times W}$, where C denotes the number of channels, and H and W denote the height and width, respectively. A simple definition of graph convolution [57] can be expressed as follows:

$$X_g = AXW \tag{4}$$

where A represents the adjacency matrix and W is a matrix containing filter parameters. As shown in Fig. 5(a), in SGCN, we first compress the number of channels of input features $X_{\text{in}}\in\mathbb{R}^{C\times H\times W}$ by 1×1 convolution operations $\mu(\cdot)$, $\varphi(\cdot)$, and $\omega(\cdot)$, respectively, and use reshape operation to change their dimension to $\mathbb{R}^{HW\times C/2}$. We use the defined form of spatial graph convolution learning and matrix multiplication to obtain a new feature, which can be described as follows:

$$\begin{aligned} M_{\text{sgcn}} &= A_{\text{sgcn}}XW_{\text{sgcn}} \\ &= \mu(X_{\text{in}})\varphi(X_{\text{in}})^{\text{T}}\omega(X_{\text{in}})W_{\text{sgcn}} \end{aligned} \tag{5}$$

where M_{sgcn} represents the output of SGCN, W_{sgcn} represents the weight matrix, and the superscript T represents the transpose operation. A_{sgcn} represents the adjacency matrix of SGCN, which can be represented by the matrix multiplication of $\mu(X_{\text{in}})$ and $\varphi(X_{\text{in}})^{\text{T}}$. Considering the computational complexity of matrix multiplication, we will replace $(\mu(X_{\text{in}})\varphi(X_{\text{in}})^{\text{T}})\omega(X_{\text{in}})$ with $\mu(X_{\text{in}})(\varphi(X_{\text{in}})^{\text{T}}\omega(X_{\text{in}}))$, which can reduce the computational complexity of the measurement matrix from $\text{O}((HW)^2)$ to $\text{O}(HW)$. We multiply the resulting adjacency matrix by $\mu(X_{\text{in}})$ again to obtain global spatial information. The softmax operation used in this process can improve the stability of the data and provide better convergence [58]. After this, a 1×1 1-D convolutional layer is used to realize the weighting process from the hidden layer to the output layer, and a 1×1 2-D convolutional layer is used to adjust the number of channels. Finally, we use elementwise addition to perform residual learning and obtain the final output features. This process can be described as follows:

$$X_{\text{sgcn}} = \text{Conv1}\times 1\big(M_{\text{sgcn}}\big) + X_{\text{in}}. \tag{6}$$

2) CGCN Module: Besides using SGCN to model the long-range contextual information of pixels of an image in the spatial dimension, we also introduce a CGCN to efficiently capture the interdependence among abstract features at the channel level. As shown in Fig. 5(b), we define the process of the new feature map obtained by convolution of the channel map as follows:

$$M_{\text{cgcn}} = \big(I + A_{\text{cgcn}}\big)XW_{\text{cgcn}} \tag{7}$$

where $A_{\text{cgcn}}\in\mathbb{R}^{C/8\times C/8}$ represents the adjacency matrix for diffusing information across nodes, I represents the identity matrix, and $W_{\text{cgcn}}\in\mathbb{R}^{C/4\times C/4}$ represents the trainable edge weight matrix. A_{cgcn} and W_{cgcn} are achieved by two 1-D 1×1 convolutional operations. CGCN replaces the adjacency matrix to $(I + A_{\text{cgcn}})$ for Laplacian smoothing [59], [60] to propagate node features over the graph. The identity matrix I is used to perform shortcut connections to alleviate optimization difficulties. We compress the number of channels of input features from C to $C/4$ and $C/8$ by two 1×1 2-D convolution operations $\nu(\cdot)$ and $o(\cdot)$, respectively, and use matrix multiplication of $\nu(X_{\text{in}})$ and $o(X_{\text{in}})^{\text{T}}$ to project the input feature to the channel interaction space. The obtained $M_{\text{cgcn}}\in\mathbb{R}^{C/8\times C/4}$ can be regarded as a graph structure with $C/8$ nodes, and the dimension of each node is $C/4$. This process can be redefined as follows:

$$M_{\text{cgcn}} = \big(I + A_{\text{cgcn}}\big)\nu(X_{\text{in}})o(X_{\text{in}})^{\text{T}}W_{\text{cgcn}}. \tag{8}$$

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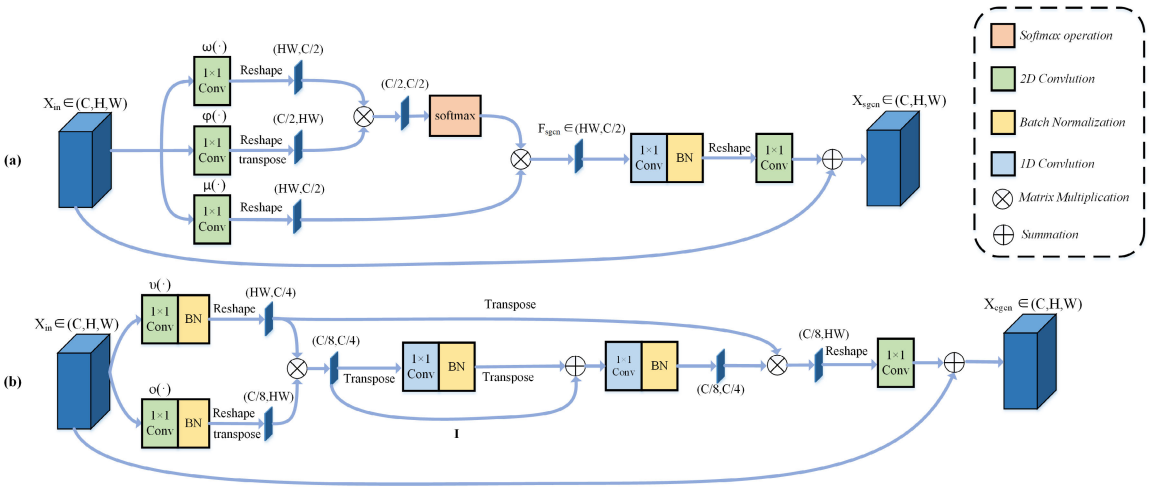


Fig. 5. Description of the DGM. (a) SGCN. (b) CGCN.

After this, we reproject M_{cgcn} by $o(X_{\text{in}})$ to the original coordinate space and use a 1×1 2-D convolution to adjust the number of channels to C. The final output result can be expressed as follows:

$$X_{\text{cgcn}} = \text{Conv1} \times 1\big(M_{\text{cgcn}} o(X_{\text{in}})\big) + X_{\text{in}}. \tag{9}$$

D. Atrous Context Self-Attention Pyramid

We designed a parallel structure ACSP based on CSA, as shown in Fig. 6. It is used to mine the spatial context of changing features and conduct self-attention learning to enhance visual representation learning. Compared with traditional self-attention based on independent query–key pairs, CSA additionally uses atrous convolution to combine input contextual information to achieve self-attention learning more efficiently. Four parallel CSAs with different dilated rates are used to capture spatial context at different scales to accommodate varying sizes of regions of interest. Specifically, given the input $F_{\text{in}} \in R^{C\times H\times W}$, we first learn the adjacency context of each spatial pixel through a 3×3 convolution with linear rectification function (ReLU) activation function. We then use four CSA branches with different dilated rates to combine contextual information at different scales and perform self-attention learning. We set their dilated rates to [1, 6, 12, 18], respectively. After that, we concatenate them in the channel dimension and use a 1×1 convolution operation to rescale their channel number to C. Finally, we add the result and the input at the element level to perform residual learning and obtain the final output $F_{\text{out}} \in R^{C\times H\times W}$. This process can be summarized as follows:

$$\begin{aligned} F' &= \text{Conv}_{3\times 3}(F_{\text{in}}) \\ F_c &= \text{Concat}\big(\zeta_{\text{dr}=1}(F'), \zeta_{\text{dr}=6}(F'), \zeta_{\text{dr}=12}(F'), \zeta_{\text{dr}=18}(F')\big) \\ F_{\text{out}} &= F_{\text{in}} + \text{Conv1} \times 1(F_c). \end{aligned} \tag{10}$$

Among them, $\zeta(\cdot)$ represents a CSA module, and dr represents the dilated rate set by it. In the implementation process of a single CSA, the way we obtain key, query, and value can be described by expressions as follows:

$$\begin{aligned} K &= F' W_k \\ Q &= F' \\ V &= F' W_v. \end{aligned} \tag{11}$$

Among them, K, Q, and V represent key map, queries, and value map, respectively, and their shapes are $R^{C\times H\times W}$. The embedding matrices W_k and W_v are learned through an atrous convolution with a kernel size of k (we set k to 3) and a dilated rate of dr and a 1×1 convolution layer, respectively. Key map reflects the static local context information restricted by dr through a $k \times k$ atrous convolution. We then splice the key map and queries in the channel dimension through the concatenation operation and obtain a dynamic multihead self-attention matrix $M_{kq} \in R^{H\times W\times(k\times k\times C_h)}$. C_h represents the number of headers, which we set to 8. This process can be expressed as follows:

$$M_{kq} = \text{Concat}(K, Q) W_\alpha W_\beta. \tag{12}$$

Among them, W_α and W_β represent two 1×1 convolution parameter matrices with ReLU activation function and BatchNorm. For each head of the self-attention matrix M_{kq}, its local attention matrix of size $k \times k$ at each spatial position is learned using query and key combined with contextual information instead of independent query–key pairs. This approach effectively enhances self-attention learning by exploring the static context of the input. After this, CSA aggregates the value map through local matrix multiplication [61], [62] and the obtained attention matrix

$$F_{qkv} = \circledast\big(M_{kq}, V\big). \tag{13}$$

Among them, $\circledast()$ represents local matrix multiplication, which performs matrix multiplication of C_h $k \times k$ local attention matrices of each spatial position of M_{kq} with the values of the $k \times k$ grid in the value map. $F_{qkv} \in R^{C\times H\times W}$ is represented as attended feature map, which connects the aggregated features of all heads and captures the dynamic contextual information in the input features. CSA then performs concatenation operations on the obtained F_{qkv} and key map to fuse static and dynamic context information and uses a channel attention [63] to obtain the final output. This process can be

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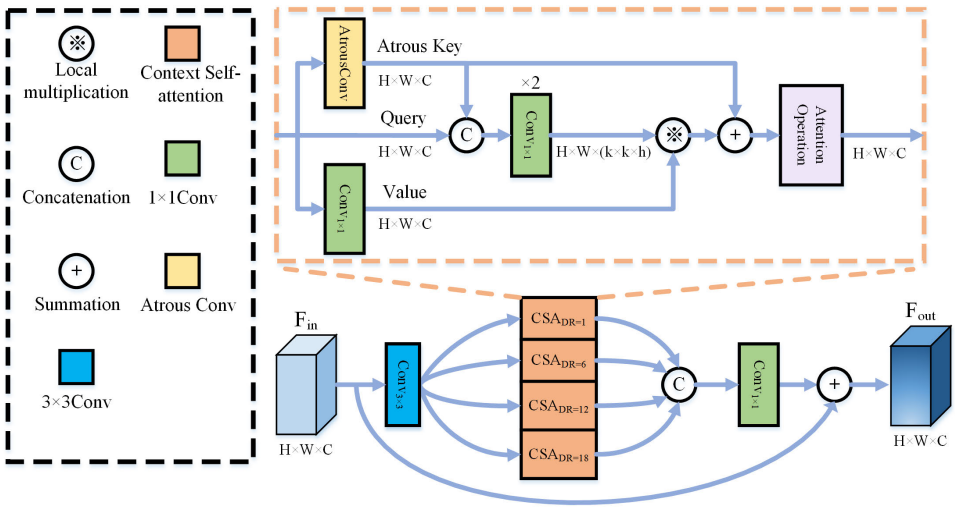


Fig. 6. Illustration of the ACSP, which contains four parallel contextual self-attention branches with different dilate rates (DRs).

expressed as follows:

$$\begin{aligned} F_{\text{fus}} &= \text{Sum}\big(\text{Concat}\big(F_{qkv}, K\big)\big) \\ \gamma &= \text{Softmax}(\rho(\tau(\text{Mean}(F_{\text{fus}})))) \\ F_{\text{csa}} &= \gamma * F_{\text{fus}}. \end{aligned} \tag{14}$$

Among them, $\rho(\cdot)$ and $\tau(\cdot)$ represent two 1×1 convolutional layers with ReLU activation function, γ represents the weight matrix, and $F_{\text{csa}} \in R^{C\times H\times W}$ is the output of a CSA branch.

E. Other Network Details

1) Loss Function: We use the minimized cross-entropy loss to optimize the model during the training phase, which is formally defined as follows:

$$L = \frac{1}{H_0 \times W_0} \sum_{h=1,w=1}^{H,W} l(P_{hw}, Y_{hw}) \tag{15}$$

where H_0 and W_0 represent the height and width of the original image, respectively, $l(P_{hw}, y) = \log(P_{hwy})$ is the cross-entropy loss, and Y_{hw} is the label of the pixel at (h, w) position.

2) Feature Difference Module: We introduced a feature difference module in the decoder to effectively calculate the difference between the feature maps of the prechange and postchange images. Two refined features obtained from the Siamese encoder are first fused by concatenation. Then, a convolutional layer with a kernel size of 3×3 and a padding of 1 is used to reduce the number of channels in the fused feature maps by half. Finally, we apply the ReLU activation function and use batch normalization for normalization to obtain the change-discriminating features. The process of the feature difference module can be formally defined as follows:

$$X_d = B\Big(R\Big(\text{Conv3}\times 3\Big(\text{Concat}\Big(X'_{\text{pre}}, X'_{\text{post}}\Big)\Big)\Big)\Big) \tag{16}$$

where X_d represents the result obtained by the difference module, X'_{pre} represents the intermediate features obtained by processing the image before the change through the feature extractor, X'_{post} represents the intermediate features obtained by processing the changed image through the feature extractor, R represents the ReLU activation function, and B represents BatchNorm.

IV. Experiments

A. Datasets

LEVIR-CD [8] is a large-scale public dataset obtained from Google Earth. It contains 637 pairs of dual-temporal RS images at a resolution of 1024×1024 pixels (0.5 m/pixel). The dataset primarily reflects land use changes in urban and suburban areas over a time span of 5–14 years, including the construction and demolition of buildings, such as apartment complexes, small garages, and large warehouses. These dual-temporal images are annotated using binary labels (0 indicating no change and 1 indicating change). To provide clearer visualization of detection results and enhance detection efficiency, we randomly crop the 637 pairs of images into nonoverlapping 256×256 resolution images and divide them into test/train/validation sets at a ratio of 2:7:1.

WHU-CD [64] is a dataset that records urban reconstruction after a 6.3 magnitude earthquake in February 2011. The dataset consists of 16 077 buildings in a 20.5-km^2 area captured in aerial photographs with a resolution of $32\,507 \times 15\,354$ pixels (0.075 m/pixel). We also obtain 6096/762/762 pairs of 256×256 resolution subimages for training/validation/testing through random cropping and image enhancement.

DSIFN-CD [23] is a public dataset obtained from Google Maps (2 m/pixel), covering six different cities in China, including Beijing, Chengdu, Chongqing, Shenzhen, Xi'an, and Wuhan. The paired dual-temporal images in this dataset mainly describe changes in land cover, such as roads, farmland, and buildings. We first obtain 3940 pairs of 512×512 resolution subimages for the training and validation sets through cropping and image enhancement of all cities except Xi'an. We also crop 48 pairs of subimages for testing from the Xi'an images.

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B. Implementation Details and Evaluation Metrics

1) Implementation Details: We built our model using PyTorch version 1.7 and trained it on an NVIDIA RTX TITAN GPU with 24-GB memory. To improve the model's generalization ability, we applied data augmentation techniques, including random flips, Gaussian blur, random color jitter, random scaling, and random cropping to the input images. We used AdamW as the optimizer with a weight decay of 0.01 and beta values of (0.9, 0.999). The training consisted of 200 epochs with a batch size of 12, and the initial learning rate was set to 0.0001, which was linearly decayed to 0 during the training process. The training dataset consisted of three datasets, namely, LEVIR-CD, DSIFN-CD, and WHU-CD.

2) Evaluation Metrics: In comparison with other advanced methods, we chose $F1$ score and intersection over union (IoU) of the change category as the primary evaluation metrics. In addition, we also used precision, recall, and overall accuracy (OA) as additional evaluation metrics. The definitions of these metrics are as follows:

$$\begin{aligned} F1 &= 2\frac{\text{Precision}\cdot\text{Recall}}{\text{Precision}+\text{Recall}} \\ \text{Precision} &= \frac{\text{TP}}{\text{TP}+\text{FP}} \\ \text{Recall} &= \frac{\text{TP}}{\text{TP}+\text{FN}} \\ \text{IoU} &= \frac{\text{TP}}{\text{TP}+\text{FN}+\text{FP}} \\ \text{OA} &= \frac{\text{TP}+\text{pn}}{\text{TP}+\text{TN}+\text{FN}+\text{FP}} \end{aligned} \tag{17}$$

where TP indicates the number of true positive, TN indicates the number of true negative, FN indicates the number of false negative, and FP indicates the number of false positive.

C. Comparison With State-of-the-Art Methods

We validate the effectiveness of our method by qualitative and quantitative comparisons with nine state-of-the-art methods, including three pure CNN-based methods: fully convolutional early fusion (FC-EF) [29], FC-Siam-Di [18], and FC-Siam-Conc [29]. Three CNN and attention-based methods: dual-task constrained deep siamese convolutional network (DTCDSCN) [22], image fusion network (IFNet) [23], and Siamese network and NestedUNet (SNUNet) [42]. Three transformer-based methods: BiT [28], ChangeFormer [45], and CropLand [30].

1) *FC-EF [29]:* A network based on FCN and UNet structures. It first concatenates two input images and then uses a single branch of convolution processing to get the final result.
2) *FC-Siam-Conc [29]:* The extended network of FC-EF adopts Siamese structure to share weights and uses feature connection to fuse bitemporal images.
3) *FC-Siam-Di [29]:* The extended network of FC-EF. The only difference from FC-Siam-Conc is that it uses feature differences to fuse bitemporal images.
4) *DTCDSCN [22]:* A method based on dual attention and FCN, which adds spatial attention and channel attention to the Siamese FCN structure to obtain more salient discriminative features.

TABLE I
QUANTITATIVE COMPARISON OF OUR PROPOSED METHOD WITH OTHER STATE-OF-THE-ART METHODS ON THE LEVIR-CD DATASET, THE BEST ONES ARE MARKED IN BOLD. ALL VALUES ARE REPORTED IN PERCENTAGE (%)

Method	LEVIR-CD				
	Precision	Recall	F1	IoU	OA
FC-EF	86.91	80.17	83.40	71.53	98.39
FC-Siam-Di	89.53	83.31	86.31	75.92	98.67
FC-Siam-Conc	91.99	76.77	83.69	71.96	98.49
DTCDSCN	88.53	86.83	87.67	78.05	98.77
IFNet	**94.02**	82.93	88.13	78.77	98.87
SNUNet	89.18	87.17	88.16	78.83	98.82
BiT	89.24	89.37	89.31	80.68	98.92
ChangeFormer	92.05	88.80	90.40	82.48	99.04
CropLand	89.79	87.57	88.67	79.64	98.86
MAFGNet(Ours)	92.59	**90.80**	**91.70**	**84.64**	**99.16**

5) *IFNet [23]:* A multiscale feature fusion method based on attention and FCN, which utilizes the attention module to effectively fuse the differential features of bitemporal images and achieves deep supervision by introducing a CD loss.
6) *SNUNet [42]:* It is a network based on the NestedUNet [65] structure, which uses dense connections to fuse multilevel features and uses channel attention for deep supervision.
7) *BiT [28]:* A transformer-based approach that abstracts the input image into semantic tags containing high-level concepts of regions of interest and leverages transformers to model context.
8) *ChangeFormer [45]:* Transformer-based multiscale feature connection method, which also combines an MLP decoder.
9) *CropLand [30]:* A CNN- and transformer-based approach that aggregates context information of multiscale features through three token encoders and token decoders built on the transformer structure.

1) Experimental Results on LEVIR-CD: The quantitative experimental results on LEVIR-CD are shown in Table I. Among the pixel-based indicators, MAFGNet has achieved the best performance except precision. Specifically, MAFGNet achieved 91.70%, 84.64%, and 99.16% on $F1$, IoU, and OA, respectively, which are 1.30%, 2.16%, and 0.12% higher than the second-ranked ChangeFormer, respectively. On the recall indicator, MAFGNet achieved 90.80%, 1.43% higher than the 89.37% of the second-ranked BiT. The data in Table I demonstrate that our method outperforms other state-of-the-art methods in terms of overall CD accuracy. This may be because DGM can explore the long-range interrelationships between different objects in the spatial dimension and channel dimension after projecting the change feature into the graph space to achieve effective context modeling, which can effectively enhance the representation of the original features. Furthermore, we use simple convolutional blocks in the encoder part instead of complex network backbones, such as ResNet [14] and DenseNet [66], to extract multiscale features. We demonstrate that in multiscale network structures, even

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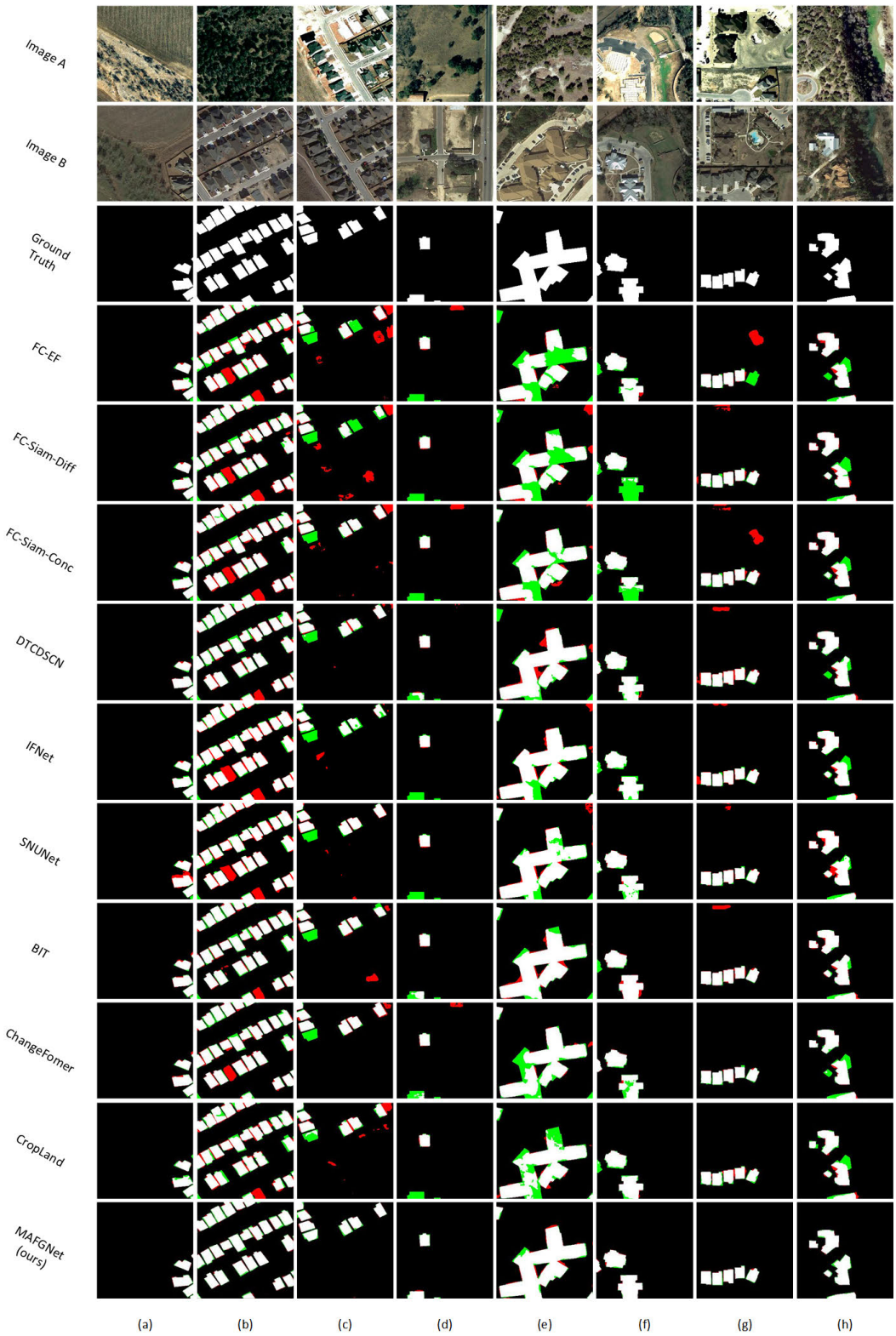


Fig. 7. Visualization of the results of our method compared with other methods on LEVIR-CD. For easier visualization, different colors are used to explain the variation, where white indicates TPs, black indicates TNs, red indicates FPs, and green indicates FNs. (a)–(h) Visual comparison results on different images selected from the test set of LEVIR-CD.

with simple feature extraction methods, improving the fusion efficiency between features at different scales can achieve effective CD performance. Fig. 7 shows a visual comparison of MAFGNet and other state-of-the-art methods on LEVIR-CD for qualitative experiments. We used four colors to better observe the changes, where black means TN, white means TP, green means FN, and red means FP. From Fig. 7(a)–(h), it can be seen that our method can significantly reduce the FN and FP parts and has a good performance in the OA. Specifically, it can be found that our method can also effectively avoid the

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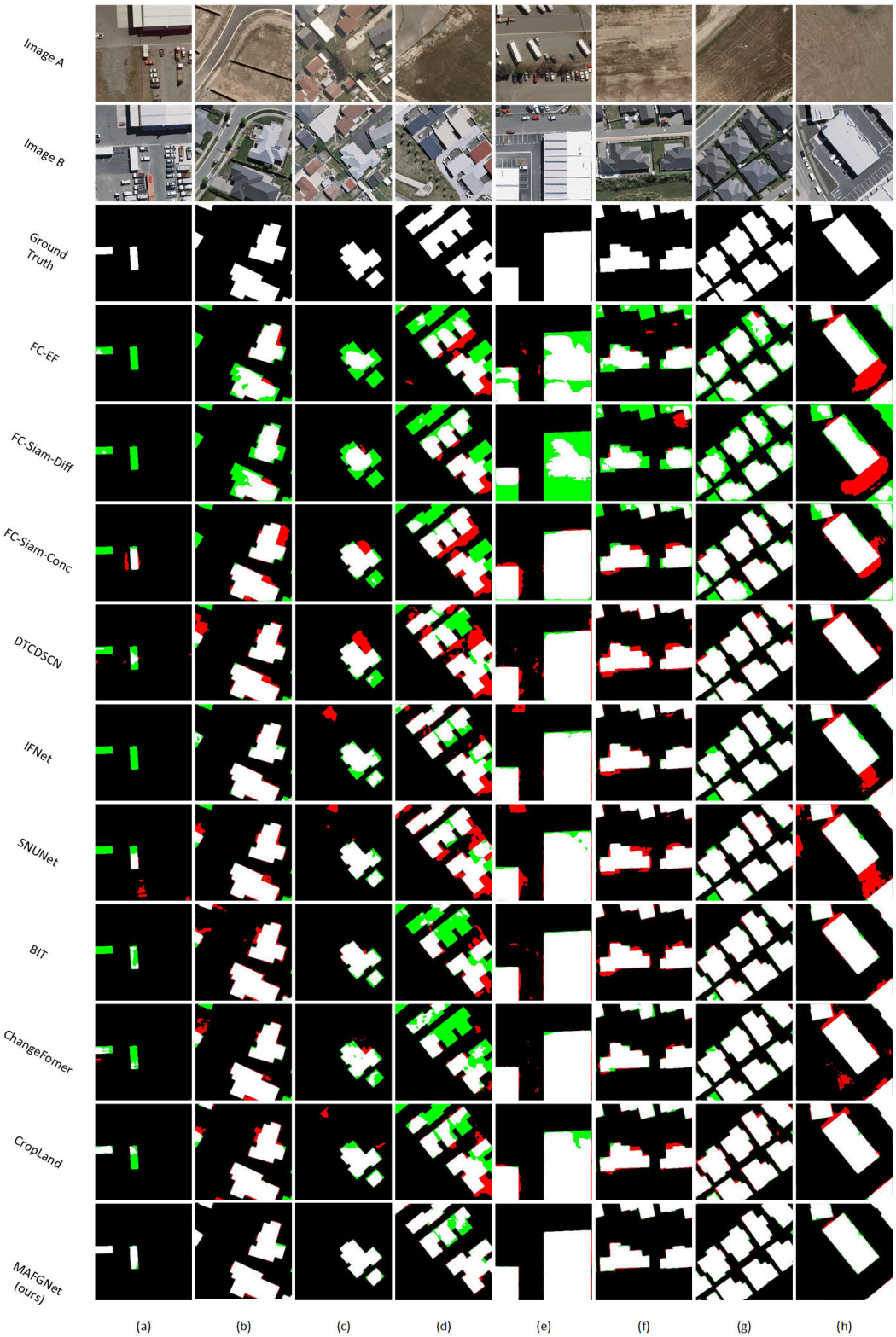


Fig. 8. Visualization of the results of our method compared with other methods on WHU-CD. For easier visualization, different colors are used to explain the variation, where white indicates TPs, black indicates TNs, red indicates FPs, and green indicates FNs. (a)–(h) Visual comparison results on different images selected from the test set of WHU-CD.

influence of shadows and environments [e.g., Fig. 7(b)–(d)]. Taking Fig. 7(d) as an example, due to the influence of the house color and the surrounding environment, it is difficult for other methods to distinguish the disappearance of the two buildings at the bottom left of the image, resulting in the lack of detection targets (the view is green). MAFGNet can reduce this omission and accurately locate the changes of interest. In addition, Fig. 7(e), (f), and (h) can prove that our method has more obvious advantages in constructing edge details of the changing image, which reflects the advantage of using

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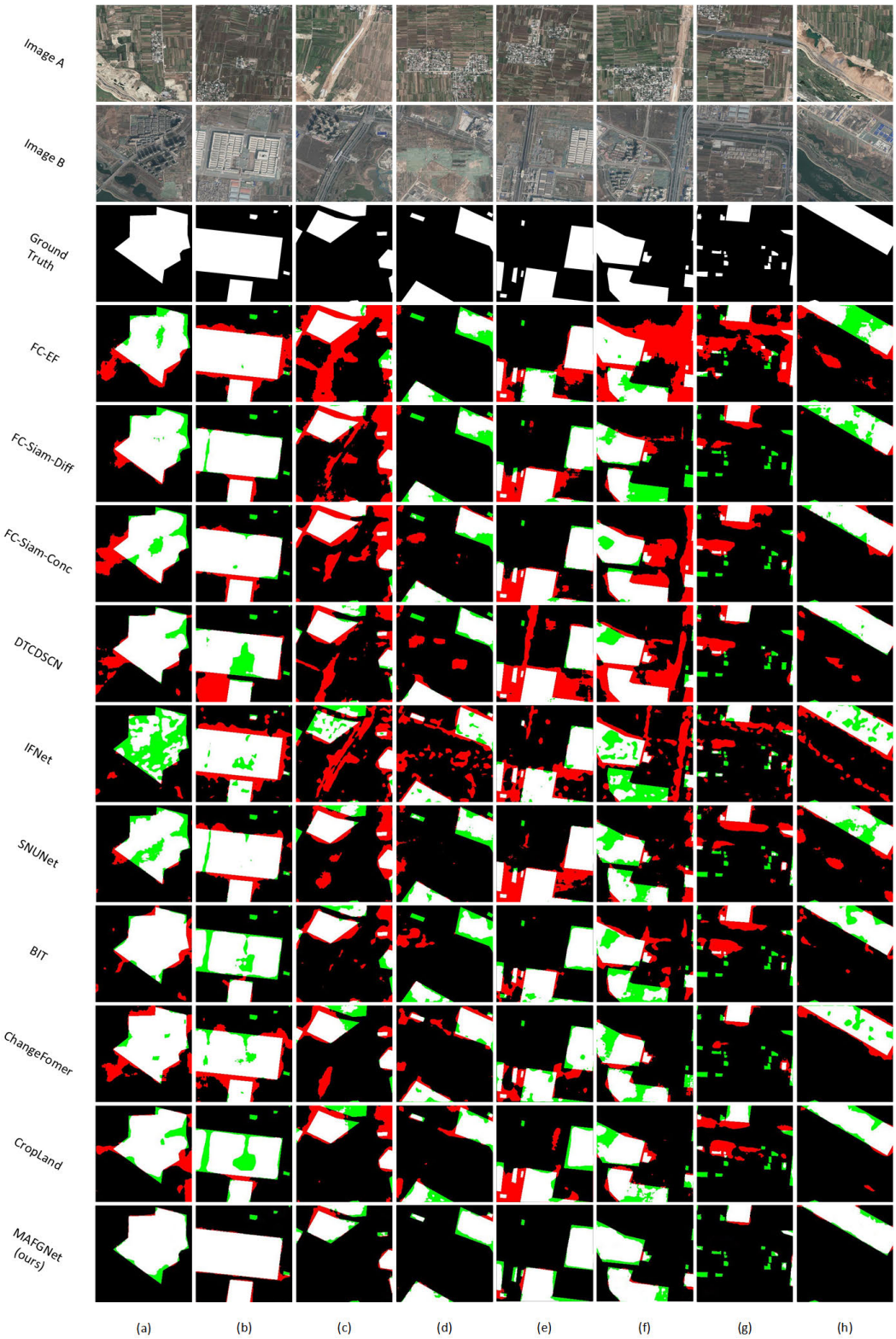


Fig. 9. Visualization of the comparison results of our method and other methods on DSIFN-CD. For easier visualization, different colors are used to explain the variation, where white indicates TPs, black indicates TNs, red indicates FPs, and green indicates FNs. (a)–(h) Visual comparison results on different images selected from the test set of DSIFN-CD.

attention to achieve multiscale fusion. These results validate that MAFGNet has superior performance on LEVIR-CD than other state-of-the-art methods.

2) Experimental Results on WHU-CD: Table II shows the comparative results of quantitative experiments obtained on WHU-CD between MAFGNet and other methods. It can be seen that our method achieves a significant improvement over other methods. MAFGNet is 5.90%/9.78%/7.95% higher than the second-ranked CropLand in terms of $F1$/IoU/recall indicators and 0.51% higher than the second-ranked IFNet

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TABLE II
QUANTITATIVE COMPARISON OF OUR PROPOSED METHOD WITH OTHER STATE-OF-THE-ART METHODS ON THE WHU-CD DATASET, THE BEST ONES ARE MARKED IN BOLD. ALL VALUES ARE REPORTED IN PERCENTAGE (%)

Method	WHU-CD				
	Precision	Recall	F1	IoU	OA
FC-EF	71.63	67.25	69.37	53.11	97.61
FC-Siam-Di	47.33	77.66	58.81	41.66	95.63
FC-Siam-Conc	60.88	73.58	66.63	49.95	97.04
DTCDSCN	63.92	82.30	71.95	56.19	97.42
IFNet	**96.91**	73.19	83.40	71.52	98.83
SNUNet	85.60	81.49	83.50	71.67	98.71
BiT	86.64	81.48	83.98	72.39	98.75
ChangeFormer	90.50	79.61	84.51	73.18	98.59
CropLand	90.81	83.79	87.16	77.24	98.81
MAFGNet(Ours)	94.42	**91.74**	**93.06**	**87.02**	**99.34**

TABLE III
QUANTITATIVE COMPARISON OF OUR PROPOSED METHOD WITH OTHER STATE-OF-THE-ART METHODS ON THE DSIFN-CD DATASET, THE BEST ONES ARE MARKED IN BOLD. ALL VALUES ARE REPORTED IN PERCENTAGE (%)

Method	DSIFN-CD				
	Precision	Recall	F1	IoU	OA
FC-EF	72.61	52.73	61.09	43.98	88.59
FC-Siam-Di	59.67	65.71	62.54	45.50	86.63
FC-Siam-Conc	66.45	54.21	59.71	42.56	87.57
DTCDSCN	53.87	77.99	63.72	46.76	84.91
IFNet	67.86	53.94	60.10	42.96	87.83
SNUNet	60.60	72.89	66.18	49.45	87.34
BiT	68.36	70.18	69.26	52.97	89.41
ChangeFormer	74.01	76.45	75.10	62.76	85.22
CropLand	71.19	79.12	73.58	60.45	82.07
MAFGNet(Ours)	**82.27**	**80.73**	**81.46**	**70.60**	**89.79**

in terms of OA indicators. Fig. 8 shows the visualization results of comparative experiments of our method and other methods. The WHU-CD dataset contains changes in dense urban buildings, and the shapes of these building objects are not regular. We selected several visualization images with different object shapes and distributions as comparison results to demonstrate the effectiveness of our model. Taking Fig. 8(d) as an example, MAFGNet presents better detection performance for irregularly changing targets and does not ignore the inconspicuous target in the top-left corner. Fig. 8(e), (f), and (h) shows that our method has good coping ability when dealing with objects with simple and complex shapes, which can effectively strengthen the CD accuracy of the model.

3) Experimental Results on DSIFN-CD: Table III shows the quantitative experimental comparison results of MAFGNet and other advanced methods on DSIFN-CD. MAFGNet achieves the best performance in all metrics. Specifically, it achieved 82.27%/81.46%/70.60% in precision/$F1$ score/IoU indicators, which are 8.26%/6.36%/7.84% higher than ChangeFormer. In the scores of recall and OA, MAFGNet is 1.61% and 0.38% higher than CropLand and BiT, respectively. Fig. 9 shows a visual comparison of our method and other methods on DSIFN-CD. It can be seen that the FN and FP parts of our method are relatively small, and the overall effect is better. Compared with the other two datasets, DSIFN-CD has lower spatial resolution, which also leads to lower indicators obtained by these methods, but our method still achieves the best detection results. This also proves that our method can adapt to different resolution datasets and is robust.

TABLE IV
COMPARISON OF FLOPS AND PARAMS OF OUR PROPOSED METHOD WITH OTHER STATE-OF-THE-ART METHODS

Method	FLOPs(G)	Params(M)
FC-EF	3.57	1.35
FC-Siam-Di	4.72	1.35
FC-Siam-Conc	5.32	1.55
DTCDSCN	13.21	31.26
IFNet	82.26	50.44
SNUNet	54.82	12.03
BiT	10.59	3.49
ChangeFormer	21.18	29.73
CropLand	14.77	16.42
MAFGNet(Ours)	62.18	6.58

In the comparison of Params and FLOPs, MAFGNet achieved a moderate and balanced result compared with other advanced methods, as shown in Table IV. In order to view the detection results in detail, we show more details of the comparison images of MAFGNet and other methods on LEVIR-CD, WHU-CD, and DSIFN-CD in Fig. 10.

D. Ablation Experiments

In order to verify the effectiveness of different modules in MAFGNet, we conducted ablation experiments on AFM, ACSP, and DGM in three datasets, and the results of the experiments are shown in Table V.

1) Ablation on AFM: We replace all AFMs used in the designed multiscale fusion encoder with common fusion blocks to fuse multiscale features for AFM ablation. This ordinary fusion block first uses bilinear interpolation to upsample the input high-level features to the same size as the low-level features, and it uses 1×1 convolution to adjust the number of channels of the low-level features to be the same as the high-level features. This process is the same as the AFM preprocessing process, except that the reweighting of the original features using the attention mechanism is removed to ensure the fairness of the experiment. A pixel-level addition operation and a convolution operation are finally used for feature fusion and compression. Due to the loss of the attention mechanism for the correction of the spatial and channel information of multiscale features, the fusion efficiency of the model becomes low, which leads to the reduction of the CD accuracy of the model. NO. 5 in Table V reflects this phenomenon on three datasets with different spatial resolutions. Compared with the complete MAFGNet (NO. 8), the model (NO. 5) only removing AFM has a decrease of 0.54%/2.42%/4.45% and 1.16%/4.14%/5.43% in $F1$ score and IoU on three datasets,

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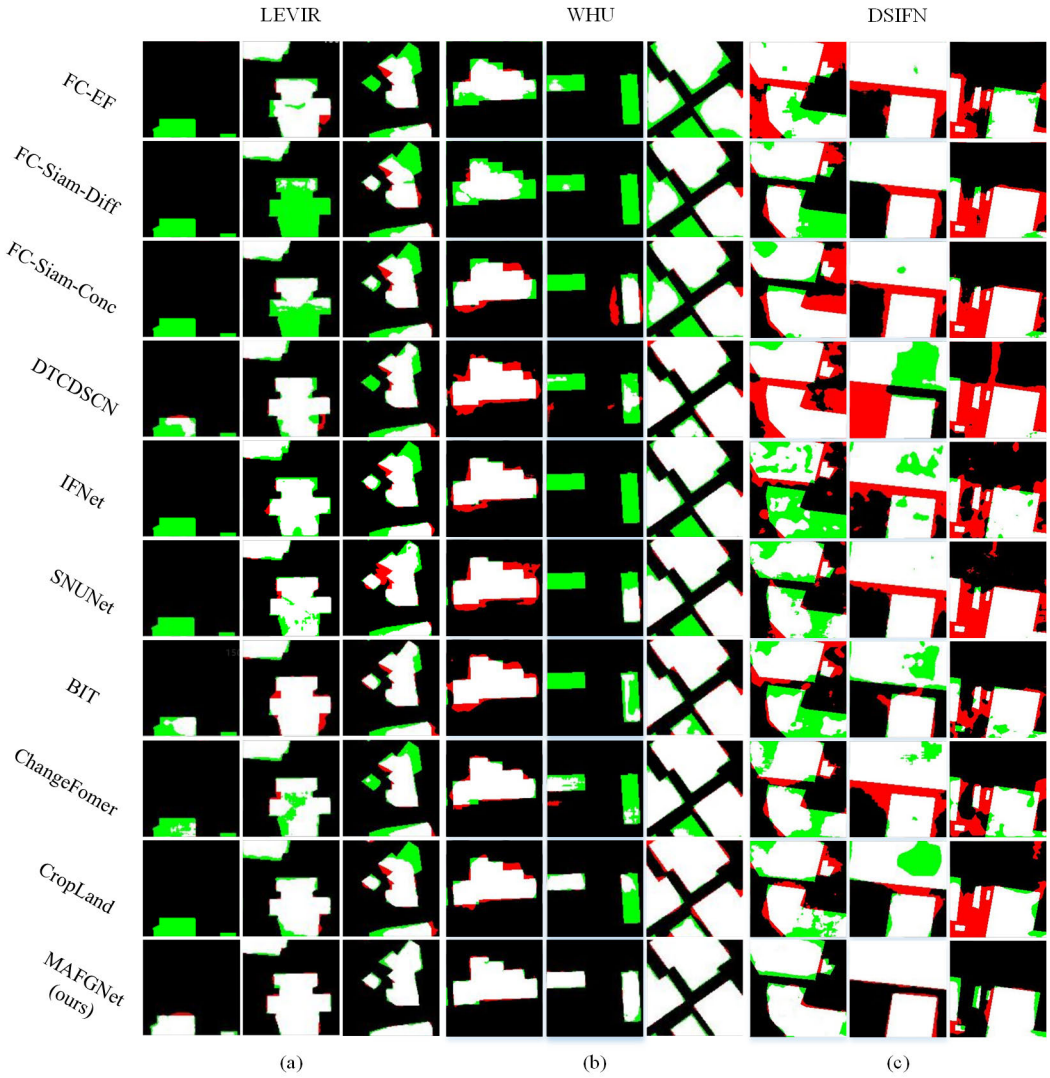


Fig. 10. Detail presentation of visualization images of different methods on three datasets. (a) Detailed image on LEVIR-CD, which corresponds to (d) and (f) of Fig. 7. (b) Detailed image on WHU-CD, which corresponds to (a) and (f) of Fig. 8. (c) Detailed image on DSIFN-CD, which corresponds to (b) and (f) of Fig. 9.

TABLE V

ABLATION EXPERIMENTS OF MAFGNET ON THREE DIFFERENT DATASETS. ABLATION WAS PERFORMED ON AFM, ACSP, AND DGM

No.	AFM	ACSP	DGM	LEVIR-CD		WHU-CD		DSIFN-CD	
				F1	IoU	F1	IoU	F1	IoU
1	×	×	×	84.49	73.14	77.38	63.11	65.14	48.26
2	✓	×	×	87.02	77.53	83.17	71.19	68.37	52.14
3	×	✓	×	86.19	75.42	82.07	69.55	68.17	51.99
4	×	×	✓	89.22	79.37	87.01	77.42	74.81	61.88
5	×	✓	✓	91.16	83.48	90.64	82.88	77.01	65.26
6	✓	×	✓	91.20	83.63	90.91	83.45	77.93	66.27
7	✓	✓	×	90.01	81.93	88.77	79.11	72.65	62.33
8	✓	✓	✓	91.70	84.64	93.06	87.02	81.46	70.60

respectively. On the other hand, when adding AFM alone (NO. 2) on the baseline (NO. 1), it increased by 2.53%/5.79%/3.23% and 4.39%/8.08%/3.88%. This proves that AFM can help the model efficiently fuse features of different scales to enhance CD performance. In addition, in order to verify the effectiveness of the multiscale attention

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TABLE VI

ABLATION EXPERIMENTS OF MULTISCALE ATTENTION FUSION LEVELS ON THE LEVIR-CD DATASET. *L*1–*L*5 REPRESENT THE LEVELS OF FUSION OF MULTISCALE FEATURES USING AFM, WHERE *L*1 ONLY USES THE RESIDUAL BLOCK TO OBTAIN DUAL-TEMPORAL FEATURES WITHOUT AFM. *F*1 SCORE, IOU, FLOPS, AND PARAMETERS ARE REPORTED

No.	LEVEL	FLOPs(G)	Params(M)	F1	IoU
1	L1	27.50	1.25	89.12	79.89
2	L2	34.21	2.02	90.01	80.16
3	L3	47.59	3.29	91.10	83.22
4	L4	62.18	6.58	91.70	84.64
5	L5	83.44	25.76	91.69	84.50

TABLE VII

ABLATION EXPERIMENTS FOR DGM. ABLATION OF SGCN AND CGCN IN DGM, WHERE NO. 1 CORRESPONDS TO NO. 7 IN TABLE V

No.	SGCN	CGCN	LEVIR-CD		WHU-CD		DSIFN-CD	
			F1	IoU	F1	IoU	F1	IoU
1	×	×	90.01	81.93	88.77	79.11	72.65	62.33
2	✓	×	91.29	83.89	91.66	83.93	76.27	65.55
3	×	✓	91.41	84.30	92.02	85.22	78.16	67.11
4	✓	✓	91.70	84.64	93.06	87.02	81.46	70.60

TABLE VIII

ABLATION EXPERIMENT OF ACSP WIDTH. CSA.W REPRESENTS THE NUMBER OF CSA BRANCHES, AND DR REPRESENTS THE DILATED RATE OF DIFFERENT CSA BRANCHES

No.	CSA.W	DR	LEVIR-CD		WHU-CD		DSIFN-CD	
			F1	IoU	F1	IoU	F1	IoU
1	1	[1]	91.39	84.18	92.59	85.73	80.58	68.66
2	2	[1,6]	91.41	84.22	92.70	85.92	80.65	68.74
3	3	[1,6,12]	91.50	84.32	92.86	86.33	81.21	70.31
4	4	[1,6,12,18]	91.70	84.64	93.06	87.02	81.46	70.60
5	5	[1,6,12,18,24]	91.60	84.51	92.93	86.77	81.22	70.50

fusion structure designed based on AFM in the encoder, we conducted quantitative verification experiments on different hierarchical structures (refer to Fig. 4), as shown in Table VI. It can be seen from the results in the table that when the level of attention fusion for multiscale features is gradually increased, the performance of the model has achieved significant growth with the increase of parameters and FLOPs. When the level reaches *L*5, the performance of the model fails to continue to grow with higher parameters and FLOPs. Therefore, we finally chose to design the structure of multiscale attention fusion as *L*4 to better balance the computational burden and performance.

2) Ablation on DGM: DGM in MAFGNet plays a vital role. The SGCN and CGCN included in it map the difference features containing change information obtained by the difference module to two different graph domain spaces. In both spaces, the long-range information of original features is aggregated, and the internal interrelationships between graph nodes are fully learned by GCN, which effectively avoids the disadvantage of CNN biased toward local information. NO. 7 and NO. 8 in Table V show the performance of MAFGNet on three datasets before and after removing DGM. After removing DGM, *F*1 score and IoU have significant reductions on these datasets, which are reduced by 1.69%/4.29%/8.81% and 2.71%/7.91%/8.27%, respectively. In addition, after adding DGM alone on the baseline, *F*1 score and IoU are improved by 4.73%/9.63%/9.67% and 6.23%/14.31%/13.62% in these datasets, respectively. These results indicate that DGM plays an important role in MAFGNet. To verify the separate importance of the two graph convolution modules in DGM, we also conduct an ablation experiment of an SGCN module and a CGCN module, as shown in Table VII. We regard the case where SGCN and CGCN are removed as the baseline, which corresponds to NO. 7 in Table V. When SGCN was used alone, the model achieved *F*1 score of 91.29%/91.66%/76.27% and the IoU of

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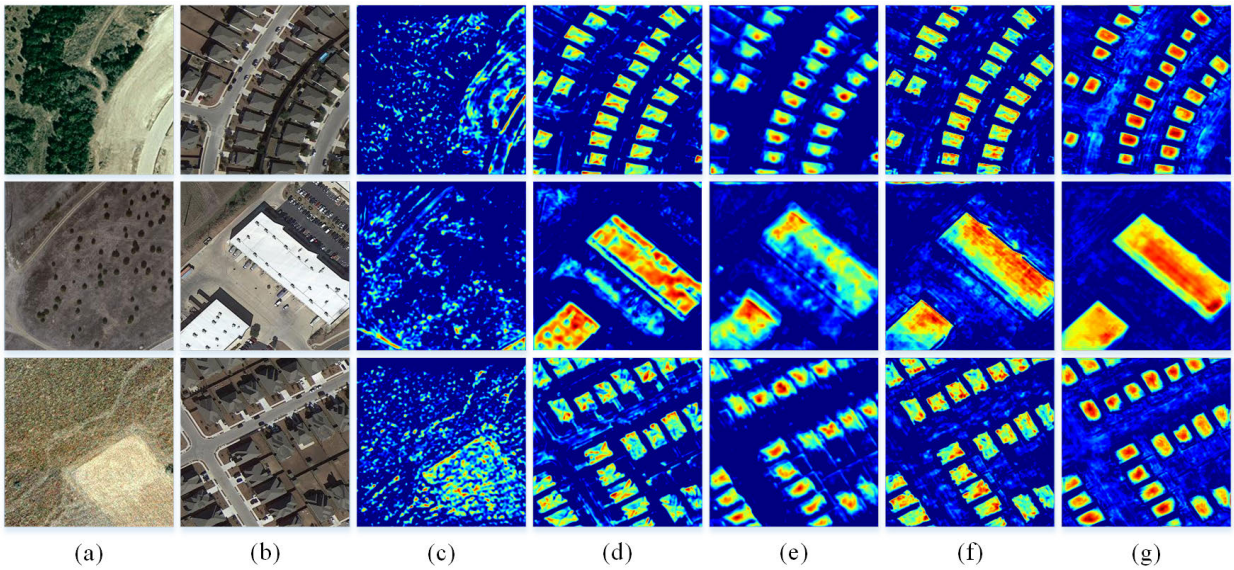


Fig. 11. Simple visualization of our network, using images from the LEVIR-CD dataset as an example. (a) and (b) Input dual-temporal images. (c) and (d) Selected high-level features obtained by stacked AFM. (e) Rough change feature obtained by the difference module. (f) and (g) Refined change feature maps obtained by ACSP and DGM, respectively. Red represents higher attention value, and blue represents lower attention value.

83.89%/83.93%/65.55% on three datasets, respectively, which were 1.28%/2.89%/3.62% and 1.96%/4.82%/3.22% higher than the baseline. On the other hand, using CGCN alone is 1.40%/3.25%/5.51% and 2.37%/6.11%/4.78% higher than the baseline, respectively. When they are mixed, the results are 1.69%/5.29%/8.81% and 2.71%/7.91%/8.27% higher than the baseline, respectively.

3) Ablation on ACSP: To demonstrate the effectiveness of our proposed ACSP module on the BCD task, we perform ablation by removing it. The ACSP module combines static and dynamic context under multiscale receptive fields from the input to achieve more efficient self-attention and feature representation. Model NO. 6 in Table V shows the $F1$ score and IoU obtained by our model without adding the ACSP module, which achieved 91.20%/90.91%/77.93% and 83.63%/83.45%/66.27% on three datasets, respectively. After adding ACSP to the model (NO. 8), $F1$ score and IoU increased by 0.50%/2.15%/3.53% and 1.01%/3.57%/4.33%, respectively. On the other hand, adding ACSP (NO. 3) alone to the baseline (NO. 1) increased 1.70%/4.69%/3.03% and 2.28%/6.44%/3.73%, respectively. In addition, Table VIII shows the analytical experimental results using different numbers of parallel CSA branches in ACSP. The results show that ACSP works best when the number of CSA branches is 4, which became our final choice. These results show that the ACSP module can effectively help the model improve CD accuracy.

E. Network Visualization

In order to more intuitively understand the role of different modules in MAFGNet, we used images from LEVIR-CD as an example to visualize different stages of the network. Fig. 11 shows these visualizations. For a given dual-temporal input [Fig. 11(a) and (b)], MAFGNet utilizes multiscale residual blocks and layer-by-layer AFMs to obtain effective dual-temporal refined features [Fig. 11(c) and (d)]. It can be seen from them that AFM can effectively capture clearer target edges, which benefits from the efficient fusion of multiscale features. Fig. 11(e) shows the change feature map obtained through the difference module. It can locate the change target, but it is rough. After multiscale self-attention learning of our proposed ACSP, we obtain a refined change feature [Fig. 11(f)], which shows effective attention to the details of the change. DGM is used to further optimize the change features after ACSP. From Fig. 11(g), we can see that after adding DGM, the model focuses more accurately on different changes of interest.

V. Conclusion

In this article, we propose a multiscale attention fusion network based on graph convolution for BCD on bitemporal images. In this network, CAFM and SAFM are frequently used to combine intermediate features of different resolutions step by step and filter out irrelevant feature information. This combination can fully explore the information of shallow features and deep features. The proposed ACSP utilizes CSA based on different dilated rates to combine the multiscale static and dynamic context of the reconstructed image to enhance self-attention learning. We then utilize SGCN and CGCN based on graph convolutional networks to capture the global spatial information of the change feature and explore the interrelationships of abstract features along the channel dimension to complete image reconstruction. We conduct qualitative and quantitative comparisons with state-of-the-art methods on different RS image datasets and also conduct ablation experiments to verify the utility of different modules.

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Yu Shangguan received the bachelor's degree from the School of Mathematics and Computer Science, Yan'an University, Yan'an, Shaanxi, China, in 2020. He is currently pursuing the master's degree with the School of Computer Science and Technology, Shandong Technology and Business University, Yantai, Shandong, China.

His research interests include computer graphics, computer vision, and image processing.



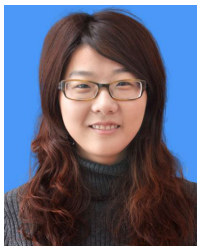
Jinjiang Li received the B.S. and M.S. degrees in computer science from the Taiyuan University of Technology, Taiyuan, China, in 2001 and 2004, respectively, and the Ph.D. degree in computer science from Shandong University, Jinan, China, in 2010.

From 2004 to 2006, he was an Assistant Research Fellow at the Institute of Computer Science and Technology, Peking University, Beijing, China. From 2012 to 2014, he was a Post-Doctoral Fellow at Tsinghua University, Beijing. He is currently a Professor at the School of Computer Science and Technology, Shandong Technology and Business University, Yantai, China. His research interests include image processing, computer graphics, computer vision, and machine learning.



Zheng Chen received the B.S. degree from Shandong Agricultural University, Tai'an, China, in 2012, the M.S. degree from Shandong Normal University, Jinan, China, in 2015, and the Ph.D. degree in signal and information processing from the Dalian University of Technology, Dalian, China, in 2022.

He is currently a Lecturer at Shandong Technology and Business University, Yantai, China. His research interests include computer vision, hand pose estimation, and hand shape recovery.



Lu Ren received the Ph.D. degree in signal and information processing from the Dalian University of Technology, Dalian, China, in 2021.

She is currently a Lecturer at Shandong Technology and Business University, Yantai, China. Her research interests include sentiment analysis and text mining.



Zhen Hua received the B.S. and M.S. degrees in electrical automation from the Taiyuan University of Technology, Taiyuan, China, in 1989 and 1992, respectively, and the Ph.D. degree in electronic information engineering from the China University of Mining and Technology, Beijing, China, in 2008.

She is currently a Professor at Shandong Technology and Business University, Yantai, China. Her research interests include computer-aided geometric design, information visualization, virtual reality, and image processing.

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